

Vishal Kumar Allam

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PROFESSIONAL SUMMARY

- Results-driven Software and Data Engineer with **5+ years of IT experience**, including **3+ years of production experience with Palantir Foundry and AIP**, delivering enterprise-scale data pipelines, Ontology models, operational applications, and AI-driven solutions across healthcare, SaaS, and enterprise environments.
- Hands-on expertise with **Palantir Foundry, Pipeline Builder, Transforms, PySpark, Polars, SQL, Python, Object Storage, Code Workbook, Code Repositories, Ontology, Workshop, Contour, Slate, Notepad, AIP Logic, and AIP Agent Studio**.
- Designed scalable **raw-to-bronze-to-silver-to-gold data architectures** ingesting structured, semi-structured, relational, and API sources while managing schema drift, data lineage, inconsistent identifiers, overlapping source systems, conflicting definitions, and observable data-quality checks.
- Extensive experience designing **Ontology schemas, object types, link types, property types, property type governance, Foundry Functions, function-backed objects, and Action Types** supporting operational business workflows and writeback capabilities.
- Built **Workshop applications, AIP Logic prompt chains, and AIP Agent workflows** grounded in Ontology objects, incorporating LLM prompting, tool calls, structured evaluation sets, measurable pass/fail criteria, agent memory, reasoning-trace analysis, and failure-mode testing.
- Strong production programming expertise in **Python and SQL** with reusable PySpark/Polars transforms, unit testing, Git-based development, CI/CD, REST APIs, AWS, Java, C#, TypeScript, Playwright, Selenium, and enterprise automation engineering.
- Experienced in translating ambiguous business problems into Foundry-solvable components through **client discovery, deskside user sessions, data analysis, stakeholder collaboration, and operational decision modeling**, identifying what users need to approve, escalate, prioritize, and act on before implementation.

EDUCATION

University of Missouri Kansas City

"Master of Science" in Computer Science

Kansas City, MO

Jan. 2023 – May 2024

Jawaharlal Nehru Technological University

"Bachelor of Technology" in Computer Science

Hyderabad, India

Aug. 2015 - May 2019

TECHNICAL SKILLS

- **Programming & Data Processing:** Python, SQL, PySpark, Polars, Pandas, Java, C#, TypeScript, JavaScript, .NET, Distributed Data Processing, Reusable Transforms, Unit Testing.
- **Palantir Foundry:** Foundry, Pipeline Builder, Transforms, Object Storage, Code Workbook, Code Repositories, Foundry Functions, Workshop, Notepad, Contour, Slate, Git-Based Workflows.
- **Ontology & Operational Applications:** Ontology Design, Ontology Schemas, Object Types, Link Types, Property Types, Property Type Governance, Function-Backed Objects, Action Types, Writeback Actions, Workshop Widgets.
- **AIP & Generative AI:** AIP Logic, AIP Agent Studio, LLM Prompting, Prompt Engineering, Prompt Chaining, Ontology-Grounded Reasoning, Agent Tasks, Tool Calls, Agent Memory, Reasoning Traces, Failure-Mode Testing, Structured Evaluation Sets, Pass/Fail Criteria.
- **Data Engineering:** ETL/ELT, Data Pipelines, Raw/Bronze/Silver/Gold Architecture, Structured & Semi-Structured Data, API Ingestion, Schema Drift, Data Lineage, Data Quality, Pipeline Observability, Data Reconciliation, Source-to-Target Mapping, Data Modeling.
- **APIs, Cloud & Databases:** REST APIs, RESTAssured, Postman, SoapUI, Swagger/OpenAPI, JSON, AWS, AWS Lambda, PostgreSQL, MySQL, Oracle, SQL Server, Vertica, Relational Databases.
- **Engineering & Delivery:** Git, GitHub, GitLab, Bitbucket, Jenkins, TeamCity, CI/CD, Selenium WebDriver, Playwright, Appium, Cucumber, TestNG, JUnit, PyTest, Jira, Confluence, Agile/Scrum, Client Discovery, Stakeholder Communication.

Software Engineer AI
PCMI LLC (Full Time)Oct 2024 – Present
San Francisco, CA

- Designed and maintained enterprise-scale **Palantir Foundry data pipelines** using Pipeline Builder, Python, PySpark, Polars, SQL, and Transforms to ingest structured, semi-structured, relational, and REST API sources into scalable operational data models.
- Architected **raw-to-bronze-to-silver-to-gold pipelines**, implementing cleansing, enrichment, reconciliation, lineage, schema drift handling, and observable data-quality checks for inconsistent identifiers, duplicate records, and conflicting source-system definitions.
- Designed and governed **Foundry Ontology schemas** containing object types, link types, property types, property governance, functions, and Action Types to model enterprise entities, relationships, and operational decision workflows.
- Developed production-quality **Python, PySpark, Polars, and SQL Transforms** within Code Workbook and Code Repositories, creating reusable shared transforms, modular utilities, automated unit tests, and Git-based code-review workflows.
- Built **Workshop applications** using configurable widgets, live Ontology objects, function-backed objects, Foundry Functions, and writeback Action Types enabling business users to approve, escalate, prioritize, and update operational records.
- Developed **AIP Logic** workflows using prompt chaining and LLM-backed reasoning grounded in Ontology objects, validating outputs through structured evaluation sets and measurable pass/fail criteria before production deployment.
- Built **AIP Agent Studio** workflows by scoping agent tasks, wiring tool calls to Foundry Functions, configuring memory, reviewing reasoning traces, and testing agent failure modes against representative enterprise datasets.
- Conducted client discovery and deskside sessions with business users to identify data availability, process gaps, decision points, and blockers, while developing reusable **pipeline templates, Ontology patterns, Workshop components, and Notepad reports** as internal accelerators.

Environments: Palantir Foundry, Pipeline Builder, Transforms, Python, PySpark, Polars, SQL, Object Storage, Ontology, Workshop, AIP Logic, AIP Agent Studio, Code Workbook, Code Repositories, Foundry Functions, Notepad, ETL/ELT, Raw/Bronze/Silver/Gold Pipelines, Schema Drift, Data Lineage, Data Quality, REST APIs, Git, TeamCity, Jira, Agile/Scrum.

Software Engineer AI
Client: CVSDec 2023 – Sept 2024
Remote, USA

- Developed enterprise healthcare data solutions using **Palantir Foundry, Pipeline Builder, Python, PySpark, Polars, SQL, and Transforms**, integrating structured healthcare datasets, semi-structured payloads, REST APIs, and relational source systems.
- Built scalable Foundry pipelines for ingestion, cleansing, normalization, enrichment, reconciliation, and publication of curated healthcare data into **Object Storage and Ontology**, with lineage and automated data-quality validation.
- Modeled healthcare entities using **Ontology object types, link types, property types, property governance, functions, and Action Types**, resolving inconsistent identifiers and relationships across overlapping source systems.
- Developed **Workshop operations-center applications** with Ontology-backed widgets, function-backed objects, filters, Foundry Functions, and writeback Action Types supporting operational approval, escalation, prioritization, and workflow decisions.
- Created production-quality **Python, PySpark, Polars, and SQL transforms** in Code Workbook and Code Repositories with reusable shared components, source-to-target validation, unit tests, and Git-based development workflows.
- Built **AIP Logic prompt chains** grounded in healthcare Ontology objects and configured AIP Agent Studio agents with tool calls, memory, reasoning-trace review, failure-mode testing, structured evaluations, and measurable production pass/fail criteria.
- Used **Contour** for exploratory analysis, **Notepad** for automated narrative reports and pipeline/model summaries, and **Slate** for customized embedded application use cases supporting technical and non-technical stakeholders.
- Partnered with product owners, developers, users, and platform engineers during discovery sessions to identify data availability, schema and business-process gaps, integration blockers, and operational requirements while integrating Foundry workflows with **AWS Lambda, REST APIs, Jenkins, and GitLab CI/CD**.

Environments: Palantir Foundry, Pipeline Builder, Transforms, Python, PySpark, Polars, SQL, Object Storage, Ontology, Workshop, Contour, Slate, Notepad, AIP Logic, AIP Agent Studio, Code Workbook, Code Repositories, Foundry Functions, Action Types, Healthcare Data, ETL/ELT, Data Quality, Data Lineage, Schema Drift, AWS Lambda, REST APIs, GitLab, Jenkins, CI/CD, C#, TypeScript, .NET, Playwright, Splunk, Agile/Scrum.

Software Engineer

Sept 2021 – Dec 2022

Infor

Hyderabad, India

- Designed and maintained scalable enterprise automation frameworks using **Java, Selenium WebDriver, Cucumber, TestNG, and Page Object Model**, creating reusable components and shared libraries for web application validation.
- Developed comprehensive **BDD feature files and Gherkin scenarios**, translating business requirements and acceptance criteria into maintainable automated functional, integration, system, and regression tests.
- Built mobile automation solutions using **Appium, Java, SeeTest, and Android Studio**, validating critical end-to-end workflows across supported mobile devices and application releases.
- Performed extensive **REST API testing using Postman and SoapUI**, validating JSON/XML payloads, authentication, HTTP methods, status codes, service contracts, integration behavior, and negative scenarios.
- Executed complex **SQL queries for database testing, data reconciliation, and source-to-target validation**, identifying inconsistent records, missing attributes, and discrepancies between application and backend layers.
- Integrated automated regression suites with **Jenkins CI/CD and Git-based workflows**, enabling continuous testing, rapid feedback, repeatable releases, and efficient collaboration through code reviews.
- Collaborated with product managers, developers, and QA engineers during Agile design and discovery sessions to clarify ambiguous requirements, identify high-risk components, integration dependencies, and potential release blockers.
- Performed root-cause analysis of application and integration failures using logs, API responses, database records, and automated evidence while tracking quality metrics, defects, regression results, and release-readiness risks.

Environments: Java, Selenium WebDriver, TestNG, Cucumber, BDD, Gherkin, Page Object Model, Appium, SeeTest, REST APIs, Postman, SoapUI, SQL, Database Testing, Data Reconciliation, Jenkins, Git, CI/CD, Jira, QTest, TestRail, Agile/Scrum, Functional Testing, Integration Testing, Regression Testing, Mobile Automation.

Associate Software Engineer

June 2019 – August 2021

Virtusa

Hyderabad, India

- Exhibited a solid 85% hands-on experience with Selenium WebDriver using Cucumber within a BDD Framework.
- Demonstrated expertise in manual testing, actively engaging in Functional Testing (80%), Integration Testing (85%), System Testing (90%), Regression Testing (85%), and Unit Testing (80%).
- Developed and executed automated testcases for mobile applications using Appium with the help of JAVA and reported defects in certain aspects.
- Conducted regression testing to ensure that new updates or changes didn't negatively impact existing functionalities.
- Possessed extensive experience in Agile Methodology and Scrum Teams, actively participating in 90% of Agile projects and Scrum teams. Additionally, contributed to 80% of tasks related to CI/CD Pipelining.
- Played a key role in the testing process, including Writing Test Cases (90%), Executing Test Cases (95%), Logging Defects (85%), Defect Tracking (80%), and Prioritizing Defects (90%).
- Prepared comprehensive test reports outlining discovered issues, resolutions, and recommendations for improvement.
- Evidenced decent knowledge, covering 75% of the technical stack in an n-tier application.

Environments: Java, J2EE, Appium, Selenium, Production testing, System Integration Testing, UAT, API Testing.

AWARDS & ACHIEVEMENTS

- 2019-11 Oracle Certified Associate, JAVA SE8 Programmer.
- ISTQB Certified Professional.
- Microsoft Technology Associate (MTA) – Database Administration Fundamentals, HTML- 5 Application Development Fundamentals.
- Awarded “Software Process Achievement” Award for my contribution towards the project in Virtusa.
- Received Customer Appreciation mails for quality releases.